

An hourly, idempotent bridge between DaisyBill and ORX — billing transactions, payments, submissions & errors, ready for Claude and reporting.

## THE EXCHANGE — HOW DATA MOVES

- n8n scheduler** — triggers MCP hourly via `POST /jobs/sync` on `daisybill.mcp.orthoxpress.com`
- MCP pulls DaisyBill** — bearer-token calls to `api.daisybill.com/v1` with `updated_at` watermark + 2h overlap
- Fan-out per bill** — detail, payments, submissions, attachments, tasks, notes & error reports
- Hand-off to DB API** — MCP never touches SQL; posts upsert package to `daisybill.api.orthoxpress.com`
- SP-only persistence** — DB API invokes stored procedures; raw JSON archived in `dbo.RawApiPayload`
- Claude consumes via MCP** — stdio or HTTP transport over protocol `2024-11-05`: tools, resources & prompts

## MCP TOOLS EXPOSED TO CLAUDE

|                                                                |                                                               |
|----------------------------------------------------------------|---------------------------------------------------------------|
| <b>trigger_sync</b><br>Kick off on-demand DaisyBill pull       | <b>get_sync_status</b><br>Progress, last watermark, errors    |
| <b>search_bills</b><br>Status, provider, date-range filters    | <b>get_bill</b><br>Full bill + child resources                |
| <b>list_billing_providers</b><br>All providers w/ sync health  | <b>get_billing_provider</b><br>Provider detail & activity     |
| <b>get_dashboard_stats</b><br>Volume, \$ posted, aging buckets | <b>test_connection</b><br>Validate DaisyBill API reachability |

## CAPABILITIES AT A GLANCE

- Incremental, Idempotent Sync**
  - ✓ Watermarked `updated_at` pulls with overlap window
  - ✓ UPSERT on natural keys — safe to replay any run
  - ✓ Exponential backoff + jitter on 429/5xx; circuit-break on outage
- Raw + Normalized Persistence**
  - ✓ Every upstream payload archived for audit & replay
  - ✓ Normalized domain tables drive reporting & the admin portal
  - ✓ PHI-aware: bodies redacted from logs, encrypted at rest & in transit
- Observability & Health**
  - ✓ `GET /healthz` liveness + readiness on MCP and DB API
  - ✓ Structured logs with correlation IDs per sync run & bill
  - ✓ Dashboard stats, per-provider sync success, row counts & latency

## DATA PASSED — DAISYBILL → ORX SQL

| ENTITY                | SOURCE ENDPOINT                  | CADENCE |
|-----------------------|----------------------------------|---------|
| Billing Providers     | /v1/billing_providers            | Daily   |
| Bills (hdr + detail)  | /v1/billing_providers/{id}/bills | Hourly  |
| Bill Payments         | /v1/bills/{id}/bill_payments     | Hourly  |
| Submissions / 837 ack | /v1/bills/{id}/bill_submissions  | Hourly  |
| Attachments (meta)    | /v1/bills/{id}/attachments       | Hourly  |
| Tasks                 | /v1/bills/{id}/tasks             | Hourly  |
| Notes                 | /v1/bills/{id}/notes             | Hourly  |
| Error Reports         | /v1/bills/{id}/error_reports     | Hourly  |

## REPORTING CAPABILITY

- Out-of-the-Box Dashboards**
  - ✓ Bills by status (Submitted, Accepted, Paid, Denied) — live counts & \$
  - ✓ Per-provider sync health — last run, row deltas, error rate
  - ✓ Payment posting trend, aging buckets, denial reason mix
- Claude-Driven Reporting (Prompts)**
  - ✓ `sync-status-report` — detailed run report w/ drill-down
  - ✓ `daily-summary` — prior-day activity & exceptions
  - ✓ `bill-analysis`, `provider-overview`, `troubleshoot-sync`
- Resources (JSON Feeds)**
  - ✓ `daisybill://sync/status`, `bills/summary`, `providers/list`
  - ✓ `daisybill://bills/{id}` — single-bill snapshot for ad-hoc Q&A
  - ✓ `daisybill://health` — system-wide posture for alerting

## OPERATIONAL SIGNALS

|                                                  |                                          |                                          |
|--------------------------------------------------|------------------------------------------|------------------------------------------|
| SYNC CADENCE<br><b>1 hr</b><br>Watermarked pulls | PAGE SIZE<br><b>25</b><br>Bills per page | CONCURRENCY<br><b>2x</b><br>Per-provider |
|--------------------------------------------------|------------------------------------------|------------------------------------------|

## PLATFORM & TRUST BOUNDARIES

- Runtime:** .NET 8, MCP protocol 2024-11-05, stdio + HTTP transport
- Persistence:** SQL Server 2022, stored-procedure-only writes (SPOnly)
- Auth:** Bearer to DaisyBill; internal JWT MCP↔DB API; TLS end-to-end
- Orchestration:** n8n schedules & triggers; Claude Code consumes MCP tools
- Host:** Windows Server 2022, IIS + Kestrel, OrthoXpress domain

## WHY IT MATTERS

- Single source of truth:** ORX SQL holds every DaisyBill bill, payment & ack — queryable, auditable, replayable
- Claude-native:** Billing Ops can ask questions in plain language and act through tools, not SQL
- Safe by design:** SP-only writes, PHI redaction, least-privilege — HIPAA & SOC 2 posture built in